# How Are Large Language Models Trained?

## From next-token prediction to reasoning

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Guest lecture for UVA CS 4774: Machine Learning (Instructor: Prof. Yanjun Qi)

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# Today: a quick look at how ChatGPT is trained

1. **Generative AI & LLMs** — what is an LLM?
2. **Language modeling** — predicting the next token
3. **Prompting & in-context learning** — learning by example
4. **Supervised fine-tuning** — following instructions
5. **RL basics** — learning from a score instead of a label
6. **Alignment** — learning from human preferences
7. **Verifiable rewards & GRPO** — learning to reason
8. **Wrap-up**

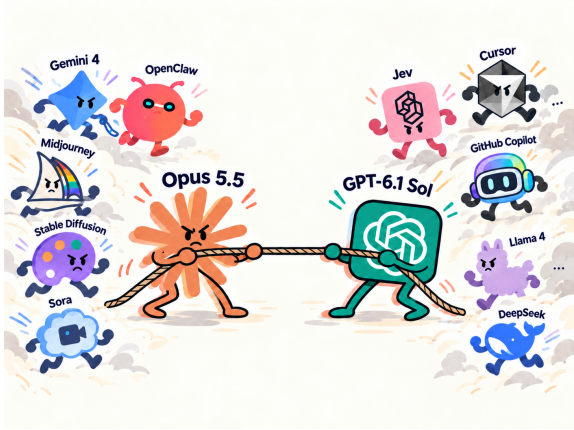
**You already know (we will reuse it):**
- logistic regression & softmax
- MLE & cross-entropy
- gradient descent / SGD
- regularization, neural nets

**Coming later in CS 4774:**
- attention & Transformers
- reinforcement learning in depth
- RAG, fine-tuning, agents

**The big picture:** every stage trains the *same* network by gradient descent on $\log p_\theta(\text{text})$. What changes from stage to stage is *which text* we use and *what weight* it gets.

# Generative AI is everywhere (2026)

- **Text & chat:** ChatGPT, Claude, Gemini, Llama, Qwen, DeepSeek
- **Code:** GitHub Copilot, Cursor, Claude Code
- **Images / video:** Stable Diffusion, Midjourney, Sora, Veo
- **Multimodal:** one model reads images, audio, and text
- **Agents:** LLMs that browse, call tools, write and run code




A new frontier model every few months.

(illustration generated with ChatGPT)

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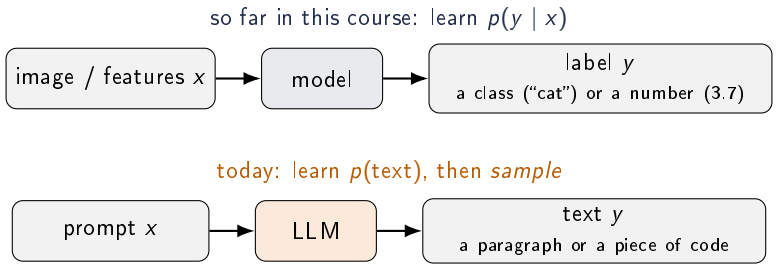
**Behind all of it:**

the **Transformer** (Vaswani et al., 2017)

+ **self-supervised pretraining**
  (on trillions of words of web text)
+ **scale**
+ **mid- and post-training**
  (fine-tuning, alignment, RL)

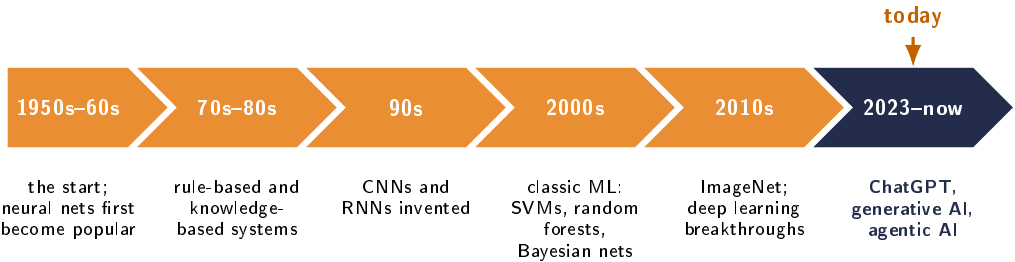
**We will focus on the last one today.**

# Discriminative vs. generative: an LLM is both



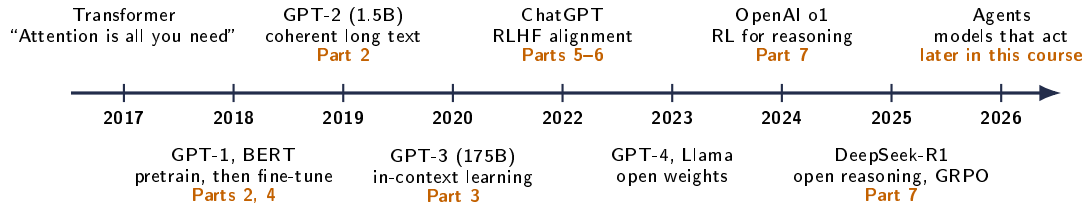

- A classifier picks one of $K$ labels.
- A generative model puts probability on *whole pieces of data* and can produce new ones.
- An LLM is *both*: a classifier over the next word, run in a loop to generate text.

# Zooming out: the machine learning timeline




**The core has not changed.** Today's models are still *neural networks trained by gradient descent on a loss*, exactly what you learned in the first half of this course. What changed is the *architecture*, the *data*, the *scale*, and the *training recipe*.

# Zooming in: 2017 to 2026




- 2018–20: **bigger** models of the same kind, trained to predict the next word.
- 2022: the same network, **post-trained** to follow instructions and human preferences.
- 2024–25: trained with **reinforcement learning** to "think" before answering: $\sim$15% $\rightarrow$ $\sim$80% on olympiad math.
- 2026: the same models as **agents**: they call tools, browse, and write code over many steps.

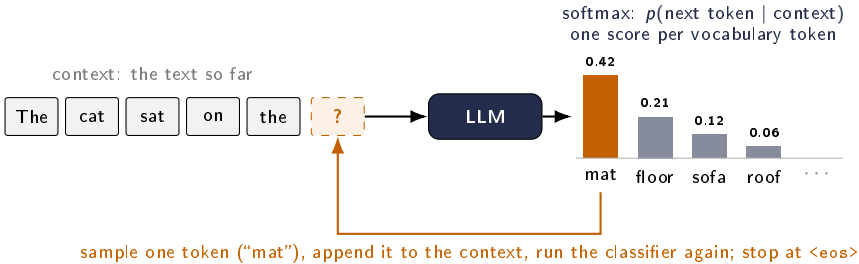
# What is a large language model?

- a **Transformer decoder** that **generates text**
- **pretrained** on trillions of words from the web
- **billions** of parameters (GPT-3: 175B, Llama-3: 405B)
- a **general-purpose** solver: translation, coding, Q&A, math

**At its core, an LLM is a classifier.**

**Input:** the text so far.

**Output:** a distribution over the *next token*, one of $|V| \approx 100{,}000$ classes.



# Base model vs. chat model: same network, different training

**Prompt**: Explain the moon landing to a 6 year old in a few sentences.

**Base model** (GPT-3, 2020)
after pre-training only: predicts the next word

Explain the theory of gravity to a 6 year old.
Explain the theory of relativity to a 6 year old in a few sentences.
Explain the big bang theory to a 6 year old.
Explain evolution to a 6 year old.

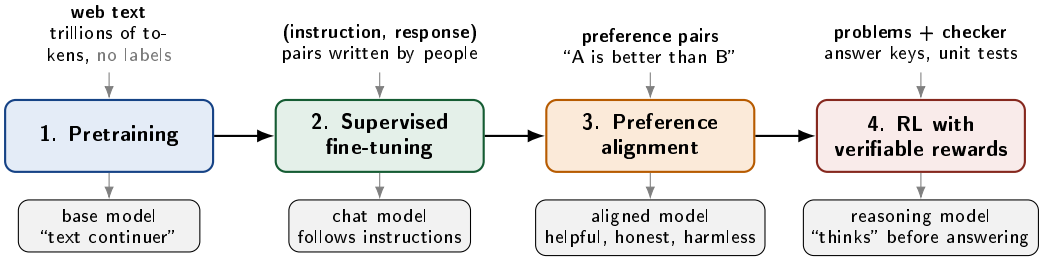
**Chat model** (InstructGPT, 2022)
the same network after post-training

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

**Same network, two training stages. Pre-training** on web text gives the base model, which continues the prompt the way web text would. **Post-training** on examples of helpful answers turns the *same network* into the chat model.

Example from OpenAI's InstructGPT announcement (2022).

# How a base model becomes a chat model: the training pipeline




- Not in the chain: **prompting / in-context learning**, which changes the *input*, not the weights.
- For each stage we ask the course's four questions: **Data? Loss? Optimization? What is learned?**

Part 2

# Language modeling

predicting the next token


Generative AI & LLMs

▶ **Language modeling**

Prompting & in-context learning

Supervised fine-tuning

RL basics

Alignment

Verifiable rewards & GRPO

Wrap-up

# A language model assigns probabilities to text

- A **language model (LM)** assigns a probability to any sequence of tokens $x_1, \ldots, x_T$.
- Chain rule of probability:

$$P(x_1, \ldots, x_T) = \prod_{t=1}^{T} P(x_t \mid x_{<t})$$

- Example: $P(\text{I eat the apple}) = P(\text{I}) \cdot P(\text{eat} \mid \text{I}) \cdot P(\text{the} \mid \text{I eat}) \cdot P(\text{apple} \mid \text{I eat the})$

So an LM only ever has to answer **one** question, over and over:
*given the text so far, what comes next?* $P(x_t \mid x_{<t})$

# Two ways to estimate $P(x_t \mid x_{<t})$

**Old way: count ($n$-gram models)**

$$P(\text{apple} \mid \text{I eat the}) \approx \frac{\#(\text{I eat the apple})}{\#(\text{I eat the})}$$

- most 4-word sequences never occur, even in a huge corpus
- fixed window, no generalization

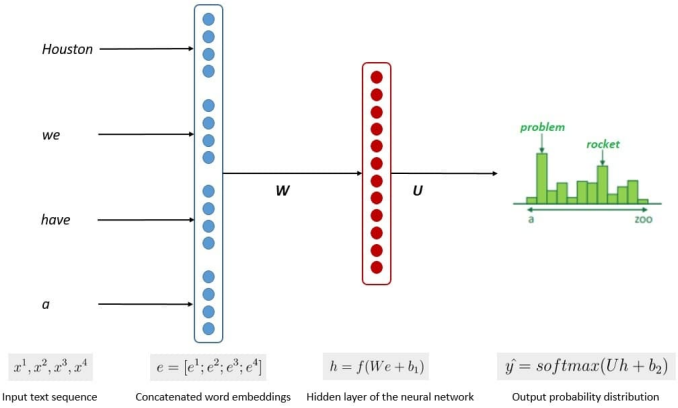
**New way: a neural network**

$$P(x_t \mid x_{<t}) \approx \mathrm{NN}_\theta(x_{<t})$$

- learns continuous representations; generalizes to unseen phrasings
- can condition on thousands of previous tokens

**You've seen this (this course):** A network with parameters $\theta$, trained by gradient descent on a loss — same as every model since Lecture 2.

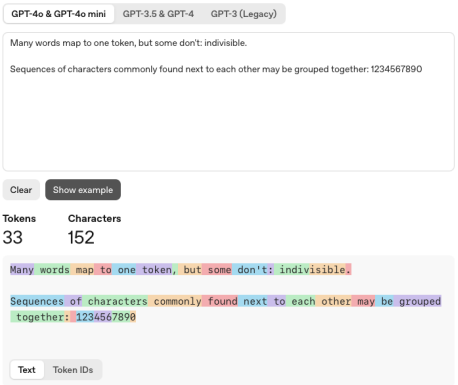
# The simplest neural language model




- Look up an **embedding** for each of the last few tokens and concatenate them; one **hidden layer**; a **softmax** over the whole vocabulary.
- This is the feed-forward network you already know, with a $|V|$-way output. Modern LLMs swap the middle for a Transformer and keep both ends.

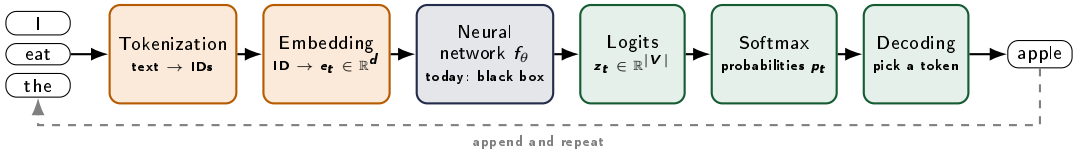
# Step 0: tokenization — text becomes a sequence of integers

- A **token** is a common chunk of characters; the **vocabulary** $V$ lists all tokens ($|V| \approx$ 50k–128k).
- **Tokenization** maps text $\rightarrow$ token IDs in $\{1, \ldots, |V|\}$.
- **Subword** tokens: frequent words are one token, rare words are split:
  `unhappiness` $\rightarrow$ `un` + `happi` + `ness`
- Foreshadowing: the model never sees letters, only tokens — one reason "how many r's in *strawberry*" is hard (Part 3).




152 characters $\rightarrow$ 33 tokens.
https://platform.openai.com/tokenizer

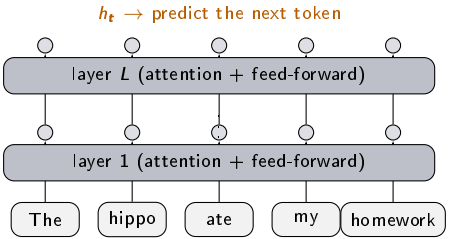
# The whole model, end to end




- **Embedding:** a learned lookup table $E \in \mathbb{R}^{|V| \times d}$; token ID $i_t$ becomes the vector $e_t = E_{i_t}$.
- **Network:** turns $e_1, \ldots, e_t$ into one vector $h_t \in \mathbb{R}^d$ that summarizes "the text so far".
- **Head:** a linear layer + softmax turns $h_t$ into a distribution over the next token.

# Inside the black box (just enough for today)

- A deep stack of layers: $h^{(\ell)} = f_\theta^{(\ell)}\big(h^{(\ell-1)}\big)$, $L = 32$ for Llama-3-8B.
- Each layer mixes information *across positions* (attention) and transforms each position (feed-forward).
- Position $t$ only sees positions $\leq t$ — no cheating when predicting $x_{t+1}$.
- All positions are processed in parallel: this is what makes training on trillions of tokens feasible.

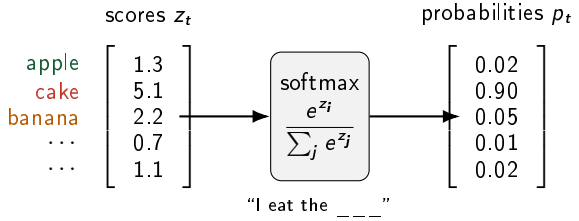



**You've seen this (10/01 NN & deep learning):** A Transformer is "just" another architecture: layers, nonlinearities, residuals, a loss at the top, gradients flowing down. Attention on 10/20, the full Transformer on 10/27.

# The output layer is multi-class logistic regression

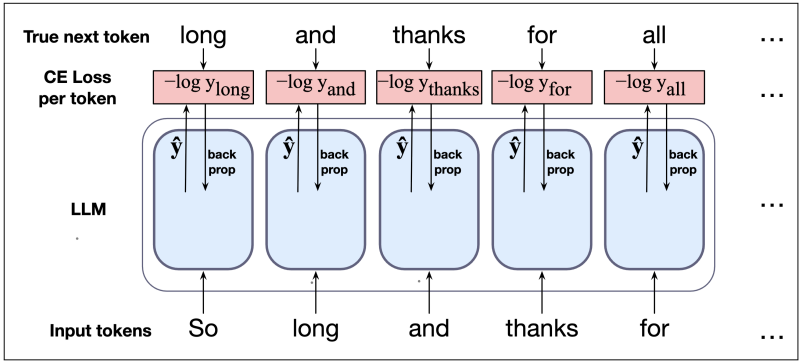
At position $t$ the network gives $h_t \in \mathbb{R}^d$. Then

$$\underbrace{z_t = W_{\text{out}}\, h_t + b}_{\text{logits: one score per token}} \in \mathbb{R}^{|V|}, \qquad p_t(i) = \frac{\exp(z_{t,i})}{\sum_{j=1}^{|V|} \exp(z_{t,j})}$$




**You've seen this (09/29 logistic regression):** $p(y{=}1 \mid x) = \sigma(w^\top x + b)$ is the 2-class case; softmax is its $K$-class generalization. Here the features are $h_t$ and $K = |V| \approx 100{,}000$. Nothing else is new.

# Pretraining in one picture



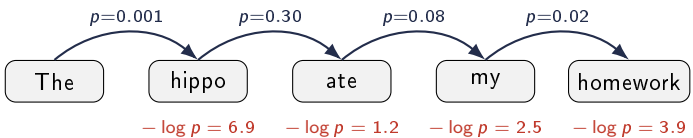

- Every position is a classification problem: the model outputs $\hat{y}$, a distribution over all possible next tokens.
- Loss at that position: $-\log$ of the probability given to the *true* next token. Average over the batch, backpropagate, update every weight (embeddings included).

Figure: Jurafsky & Martin, *Speech and Language Processing*, 3rd ed.

# Training = cross-entropy = maximum likelihood on the corpus

**Pretraining loss** (GPT-style, "causal" language modeling):

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{t=1}^{N} \log p_\theta(x_t \mid x_{<t})$$




- Target at each position = the **actual next token**; loss = − log of the probability the model gave it; average over positions.
- **Self-supervised**: the labels are free — the next word is already in the text. Minimized by **SGD (Adam)** + backprop.

**You've seen this (09/24 MLE):** $\max \prod_t p_\theta(x_t \mid x_{<t}) \Leftrightarrow \min -\sum_t \log p_\theta(x_t \mid x_{<t})$. Pretraining *is* maximum likelihood.

# Why does predicting the next word teach so much?

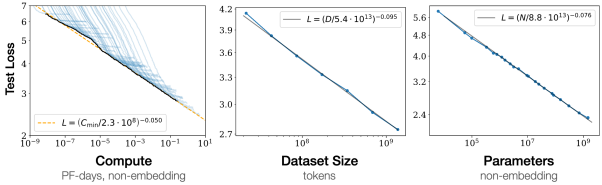
Minimizing one loss on trillions of words forces the model to pick up every regularity that helps predict the next token:

| **to fill the blank it must learn ...** | **example** |
|---|---|
| grammar | In my free time, I like to {run, banana} |
| lexical semantics | At the zoo we saw giraffes, lions, and {zebras, spoon} |
| world knowledge | The capital of Denmark is {Copenhagen, London} |
| sentiment | Edge of my seat the whole time. The movie was {good, bad} |
| translation | The word for "pretty" in Spanish is {bonita, hola} |
| arithmetic | $3 + 8 + 4 = \{15, 11\}$ |

**Pretraining is implicit multi-task learning.** Nobody wrote a sentiment loss or a translation loss; the next-token loss subsumes them.

# Scale: more of everything, predictably better

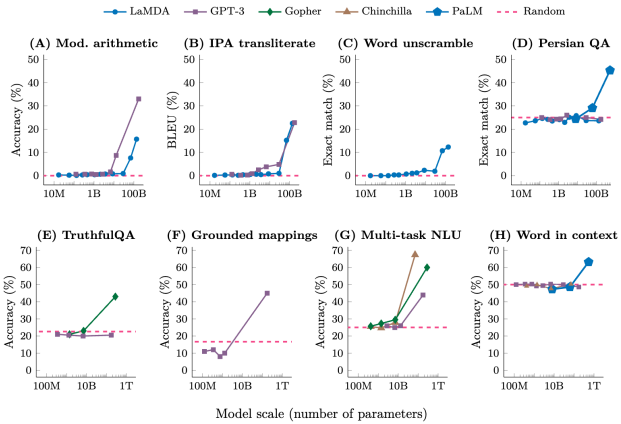
- **Data:** Llama-3 saw ∼15 trillion tokens.
- **Parameters:** 0.1B (2018) → 175B (2020) → 405B (2024).
- **Compute:** thousands of GPUs for months.
- **Scaling laws** (Kaplan et al., 2020): test loss falls as a **power law** in each factor — straight lines on log-log axes.




Test loss vs. compute, dataset size, and parameters.

**You've seen this (09/22 bias–variance):** Test loss keeps improving with more data *and* more parameters — nowhere near the regime where capacity hurts.

# Emergent abilities: some skills appear only at scale

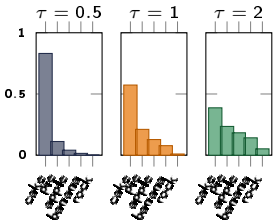



- Few-shot accuracy on arithmetic, word unscrambling, multi-task understanding, ...
- Stays at **chance** until a size threshold, then jumps — not predictable from smaller models.
- The same recipe at larger scale gave GPT-3 the ability in Part 3: learning from examples in the prompt.

# Decoding: from a probability distribution to text

Loop: compute $p_t \to$ pick a token $x_t \to$ append $\to$ repeat until <EOS>.

- **Greedy:** $x_t = \arg\max_i p_t(i)$. Deterministic; bland, repetitive.
- **Sampling:** $x_t \sim p_t$. Diverse; may pick nonsense from the tail.
- **Temperature** $\tau$: $p_t^{(\tau)}(i) \propto \exp(z_{t,i}/\tau)$ — $\tau \to 0$ greedy, $\tau \uparrow$ flatter.
- **Top-$k$ / top-$p$:** sample only from the most likely tokens.




Same logits for "I eat the ___", three temperatures.

Why ChatGPT answers the same question differently each time — and why "sample several answers to one question" is a key tool in Part 7.

Part 3

# Prompting & in-context learning

learning by example


Generative AI & LLMs
Language modeling
▶ **Prompting & in-context learning**
Supervised fine-tuning
RL basics
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Wrap-up

# A pretrained LM continues text; prompting designs the input

- A **prompt** is the text we hand the model; its only job is to continue it.
- **Prompting** = write the prompt so that the *most likely continuation* is the answer we want.
- **Zero-shot:** a task description, no examples.
- No training, no gradient, no change to $\theta$.

**Sentiment analysis by continuation**

Review: *The food was cold and the service painfully slow.*
Sentiment:

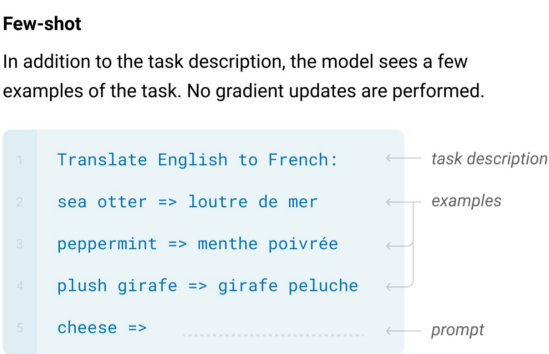
negative

**Question answering by continuation**

Q: Who wrote *Hamlet*?
A:

William Shakespeare

# In-context learning: learning from examples with frozen weights




Few-shot prompting in the GPT-3 paper (2020).

- **Few-shot / in-context learning (ICL):** put $k$ input→output examples in the prompt; the model produces the output for a new input.
- Appears at scale (GPT-3): bigger models get much better at it.
- **No parameter update.** The "learning" happens inside one forward pass.

**You've seen this (everything so far):** Every algorithm in this course learns by changing weights with gradient steps. ICL adapts *without touching the weights* — closer in spirit to KNN than to gradient descent.

# In-context learning, live: two examples change the answer

**Demo 1 — letter manipulation**

Swap the second and the penultimate letter of the following word: `pothyn`

`pohtyn` ✗

Directly answer the last one (swapping the second and the penultimate letter):
`tarehd` → `thread`, `revir` → `river`, `pothyn` →

`python` ✓

**Demo 2 — counting letters**

How many ‘r’ letters are there in the following word: “strawberry”?

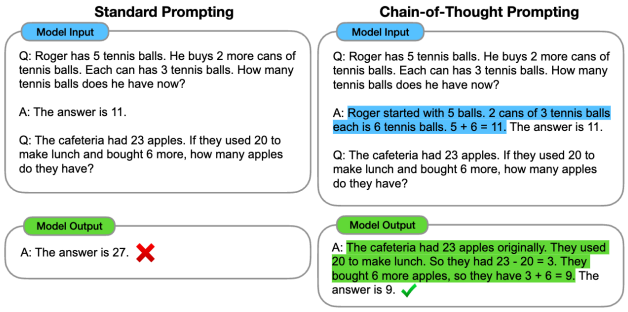
There are 2 ‘r’ letters in “strawberry”. ✗

Count how many ‘r’ letters are there in the following words:
“red”: 1, “roar”: 2, “strawberry”:

3 ✓

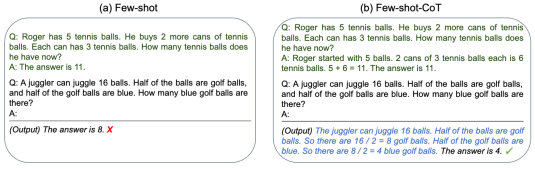
Why is this hard at all? The model sees *tokens*, not letters (Part 2) — the examples show it the pattern it needs.

# Chain-of-thought prompting: ask for the steps

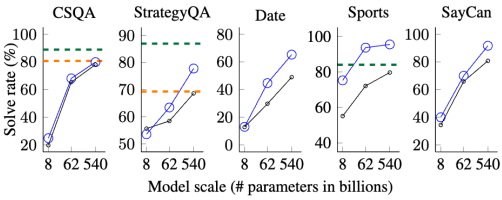



- Show (or just request) **intermediate reasoning steps** before the final answer.
- Large gains on math and logic problems.
- Why it helps: each generated step becomes context for the next prediction — the model **computes in text**.

# Chain-of-thought works zero-shot — and mostly for large models




Just append *"Let's think step by step."* (Kojima et al., 2022).




CoT helps once models are large enough; small models produce fluent but wrong chains.

Foreshadowing Part 7: reasoning models (o1, R1) learn to produce long chains of thought *on their own*, via reinforcement learning.

# The limits of prompting

- **Fragile:** results depend on wording and example choice.
- **Expensive at use time:** examples must be re-sent with every query.
- **A base model is not trying to help you:** it continues text (Part 1 demo).
- **Nothing persists:** prompting cannot permanently add a skill, a style, or a safety rule.

To change what the model does *by default*, change $\theta$.

**Part 4:** supervised fine-tuning — show it what good responses look like.

**Parts 5–7:** reinforcement learning — tell it how good its own responses were.

Part 4

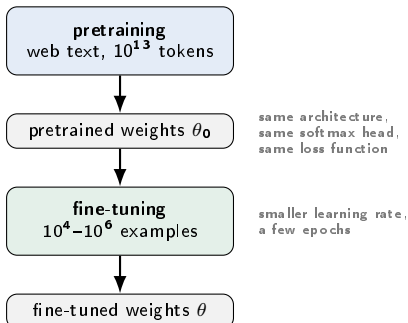
# Supervised fine-tuning

following instructions


Generative AI & LLMs
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Wrap-up

# Pretrain once, fine-tune many times

- **Pretraining:** trillions of tokens, months, tens of millions of dollars — done *once*, by a few labs.
- **Fine-tuning:** *continue training the same network* on a small, targeted dataset — hours on one GPU.
- The pretrained weights $\theta_0$ are the starting point; fine-tuning moves them a little, keeping what pretraining learned.
- Today's use: **instruction tuning** — turn the text continuer into an assistant.



# Instruction tuning: the data

**Instruction–response pairs** written or curated by people:

Label the sentiment of this sentence: *The movie wasn't that great.*

negative

Translate to Spanish: *The office was built in three months.*

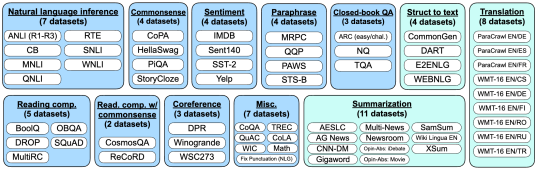
La oficina se construyó en tres meses.

**Chat format** — multi-turn, with roles, serialized with special tokens:

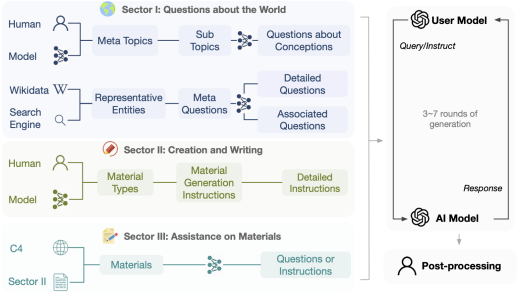
```
{"role": "user", "content": "Weather today?"}
{"role": "assistant", "content": "Sunny, high
of 75."}
{"role": "user", "content": "And tomorrow?"}
{"role": "assistant", "content": "Partly
cloudy, 72."}
```

- The model learns when to *start* and when to *stop* (<eos>), and to follow the instruction.
- $10^4$–$10^6$ examples — tiny next to $10^{13}$ pretraining tokens.

# Where instruction data comes from




**FLAN** (2022): 62 existing NLP datasets in 12 task clusters, each rephrased as natural-language instructions.



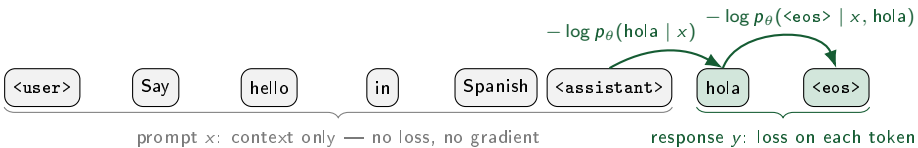

**UltraChat** (2023): millions of multi-turn dialogues generated by having one LLM play the user and another the assistant.

- **Diversity** of tasks and phrasings matters more than sheer size.
- Today much of the data is synthetic: strong models write the demonstrations, humans filter.

# SFT loss: the same cross-entropy, on the response tokens

**Supervised fine-tuning (SFT) objective** on pairs $(x, y)$ = (prompt, response):

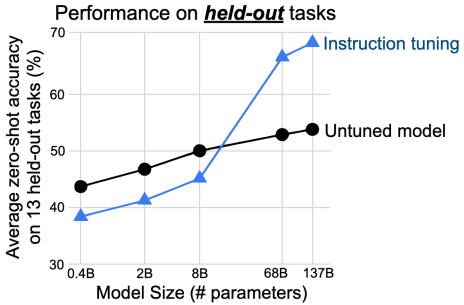
$$\mathcal{L}_{\text{SFT}}(\theta) = -\sum_{(x,y)\in\mathcal{D}} \sum_{t=1}^{|y|} \log p_\theta\big(y_t \mid x, y_{<t}\big)$$



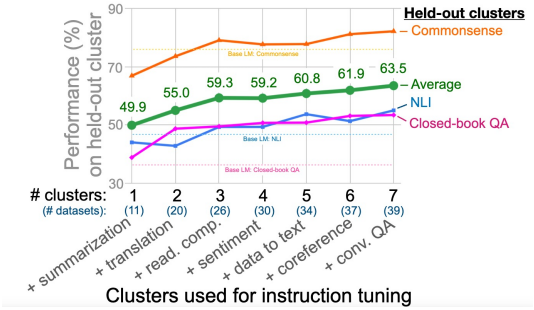

- Identical to the pretraining loss, except prompt tokens are **context only** ("loss masking").
- Same network, same softmax head, same optimizer; the model learns to **imitate** the demonstrations.

**Nothing new in the math — only new data.** What it learns: the *format and behavior* of an assistant.

# What instruction tuning buys: zero-shot generalization




Zero-shot accuracy on *held-out* task types vs. model size (FLAN).




More task clusters in the fine-tuning mix ⇒ better on unseen clusters.

- Train on 60 tasks phrased as instructions ⇒ the model follows **new** instructions with no examples in the prompt.
- Chat-style data does the same for conversation: this is how a text continuer becomes a chatbot.

# The limits of imitation

SFT = imitate human-written responses. Four problems:

1. **Costly supervision:** someone must *write* a full, high-quality response for every prompt.
2. **Every token counts the same:** the loss penalizes a typo exactly like a factual error.
3. **A ceiling:** the model can only copy the demonstrator — and for open-ended tasks there is no single "right" answer.
4. **Judging is easier than writing:** it is far easier to say *which of two answers is better*, or whether an answer is *correct*, than to write the best answer.

# A new learning paradigm: learn from evaluation, not demonstration




Let the model *generate* its own responses, *score* them, and update toward the ones that scored well. That is **reinforcement learning** (Part 5).

Two kinds of score:

- **Part 6:** "humans prefer A over B" → a learned reward model (RLHF), or used directly (DPO).
- **Part 7:** "the answer is correct" → a program that checks it (RLVR, GRPO).

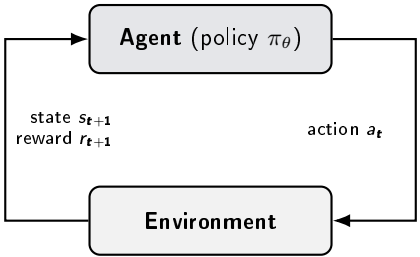
Part 5

# RL basics

learning from a score instead of a label


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# Reinforcement learning in one picture




The agent–environment loop (Sutton & Barto).

- **State $s_t$:** what the agent observes; **action $a_t$:** what it does.
- **Reward $r_t$:** a number from the environment — the only feedback.
- **Policy $\pi_\theta(a \mid s)$:** a distribution over actions given the state — a classifier, again!
- **Goal:** maximize the expected total reward $J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_t r_t\right]$.

Examples: Atari / Go (reward = score, win), a walking robot (distance), ... and a chatbot (?).
Prof. Qi covers RL properly on 10/22; today, just enough to train an LLM.

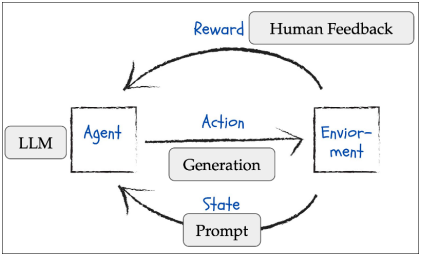
# Supervised learning vs. reinforcement learning

| | **supervised learning** (so far) | **reinforcement learning** |
|---|---|---|
| feedback | the *correct output* $y^\star$ for each input | only a *score* $r$ for the output you produced |
| data | fixed dataset, i.i.d. | generated by *your own* policy — changes as you learn |
| granularity | per example, per token | often one number at the *end* of a long sequence |
| objective | minimize loss against labels | maximize expected reward |
| hard parts | — | **exploration**, **credit assignment**, **instability** |

Why bother? For "is this helpful?" or "is this proof correct?", a **score** is far cheaper than a **demonstration**.

# The LLM as a policy

- **State** $s_t$ = (prompt $x$, tokens generated so far $y_{<t}$)
- **Action** $a_t$ = the next token $y_t \in V$
- **Policy** $\pi_\theta(a_t \mid s_t) = p_\theta(y_t \mid x, y_{<t})$ — the same softmax classifier from Part 2
- **Episode** = one complete response $y = (y_1, \ldots, y_T)$
- **Reward** $R(x, y)$ = a score for the *whole* response, given once at the end




Prompt = state, generation = action, feedback = reward.

**Objective:** $J(\theta) = \mathbb{E}_{x\sim\mathcal{D}}\,\mathbb{E}_{y\sim\pi_\theta(\cdot|x)}\big[R(x,y)\big]$

Where does $R$ come from? **Part 6**: a reward model learned from human preferences. **Part 7**: a program that checks the answer.

# Policy gradient: how to increase a reward you can only sample

We cannot backprop through $R$ (a human judgment or a black-box checker). But we can move probability mass toward responses that scored well:

$$\nabla_\theta J(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot|x)}\Big[\, R(x,y)\, \nabla_\theta \log \pi_\theta(y \mid x) \,\Big], \qquad \log \pi_\theta(y \mid x) = \sum_t \log p_\theta(y_t \mid x, y_{<t})$$

**REINFORCE** (Williams, 1992): sample a response $y$, then

$$\theta \leftarrow \theta + \alpha\, R(x,y)\, \nabla_\theta \log \pi_\theta(y \mid x).$$

- Good response ($R$ large): make every token of it more likely. Bad response: less likely.
- Needs no labels and no differentiable reward — just samples and scores.

One-line derivation: $\nabla_\theta \sum_y \pi_\theta(y) R(y) = \sum_y \pi_\theta(y)\, \nabla_\theta \log \pi_\theta(y)\, R(y)$ because $\nabla \pi = \pi\, \nabla \log \pi$.

# Read it as: sample → score → weighted SFT step

| | gradient | where $y$ comes from |
|---|---|---|
| **SFT** (Part 4) | $1 \cdot \nabla_\theta \log p_\theta(y^\star \mid x)$ | a *human* wrote $y^\star$ |
| **RL** | $R(x, y) \cdot \nabla_\theta \log \pi_\theta(y \mid x)$ | the *model* sampled $y$ |

**Policy gradient = the SFT gradient on self-generated data, weighted by how good it was.**

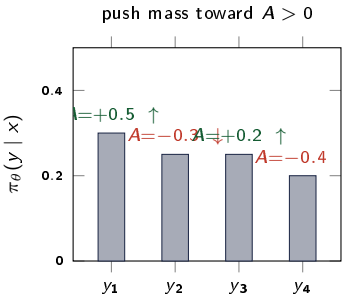
- Nothing changes in the network, the softmax head, or the optimizer.
- What changes: the data is the model's own output, and each sequence carries a *weight*.

# Baselines and the advantage: good *relative to what*?

- If every response scores 0.6–0.9, *all* get pushed up; only differences carry information, and one sample is very noisy.
- Fix: subtract a **baseline** $b(x)$ — unchanged expected gradient, much lower variance:

$$\nabla_\theta J = \mathbb{E}\big[(R(x,y) - b(x))\,\nabla_\theta \log \pi_\theta(y \mid x)\big]$$

- **Advantage** $A = R - b$: "how much better than expected".
- $b$ from a learned value function ("critic") in classic RL; from a *group of samples* in GRPO (Part 7).



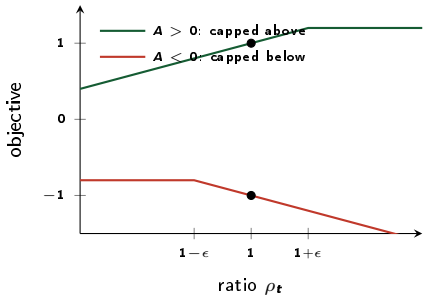

Four sampled responses to one prompt. Rewards 0.9, 0.1, 0.6, 0.0; baseline $b = 0.4$.

# Guardrail 1: don't move too far in one step (PPO clipping)

- The policy *generates its own training data*: one step too far → garbage samples → worse step. (In supervised learning the next batch corrects a bad step.)
- **PPO** (2017): per token, let $\rho_t = \dfrac{\pi_\theta(y_t \mid \cdot)}{\pi_{\text{old}}(y_t \mid \cdot)}$ and optimize

$$\min\big(\rho_t A,\ \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\, A\big).$$

- Once a token's probability has moved by $\pm\epsilon$ (e.g. 20%), it gets no further push this round.




PPO's clipped objective for one token ($|A| = 1$).

# Guardrail 2: stay close to where you started (KL penalty)

- Subtract a penalty for drifting away from a frozen **reference model** $\pi_{\text{ref}}$ (usually the SFT model):

$$\max_{\theta} \; \mathbb{E}_{y \sim \pi_\theta}\big[R(x,y)\big] \; - \; \beta \, \mathrm{KL}\big(\pi_\theta(\cdot \mid x) \,\|\, \pi_{\text{ref}}(\cdot \mid x)\big)$$

- Keeps the language fluent, prevents collapsing onto one answer, and limits exploiting flaws in the reward.
- $\beta$ trades off "get more reward" against "stay a normal language model".

**You've seen this (09/15 regularization):** $L_2$ keeps the weights $w$ near 0; the KL term keeps the *policy* $\pi_\theta$ near $\pi_{\text{ref}}$. Same idea, different space.

Part 6

# Alignment

learning from human preferences


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# Why alignment? Predicting text is not the same as being helpful

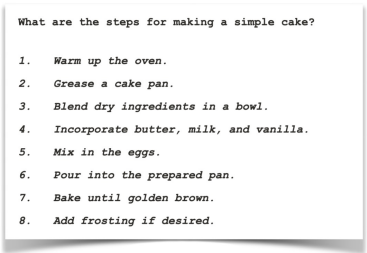
- **Objective mismatch:** "predict the next word" says nothing about being helpful, truthful, or safe.
- **Data bias:** web text contains toxic and misleading content; the LM does not distinguish good from bad behavior.
- GPT-3 (2020) was impressive but awkward to use; ChatGPT (2022) = the same kind of model + **alignment**.
- Target — "HHH": **helpful**, **honest**, **harmless**.

**How do you write a loss function for ...**

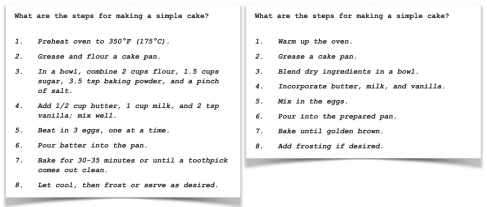
what is helpful? what is funny?
what is ethical? what is safe?

You don't. You **learn** it from human judgments — then optimize it with RL (Part 5).

# Step 1: collect comparisons, not scores



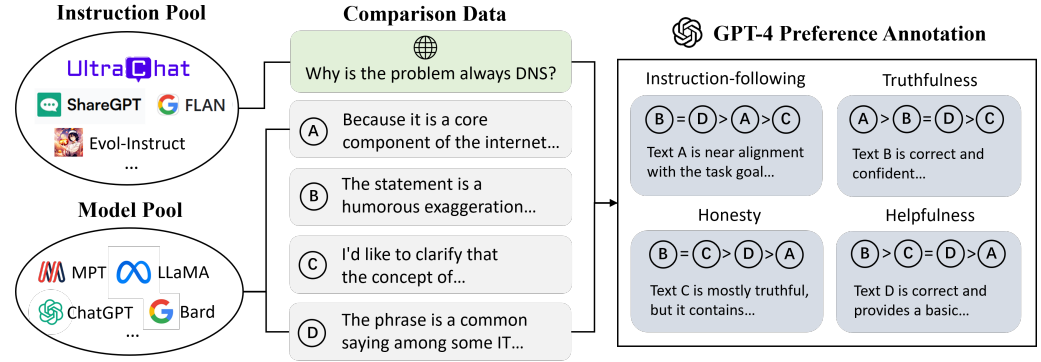

"Rate this answer 1–10": different people give very different numbers for the same text.




"Which of the two is better?": humans are much more consistent at relative judgments.

Preference data = triples $(x,\ y_w,\ y_l)$: prompt, **w**inning response, **l**osing response, both sampled from the model.

# Preference data at scale: UltraFeedback




- 60k prompts from many instruction datasets; 4 responses per prompt from different LLMs.
- Each response rated by **GPT-4** on instruction-following, truthfulness, honesty, helpfulness ("RLAIF": AI feedback instead of human feedback).
- Highest-scoring response = $y_w$; a random other one = $y_l$. This is the dataset in Colab notebook 1.

# Step 2: a reward model = logistic regression on pairs

**Bradley–Terry model.** Each response gets a hidden score $r_\phi(x, y) \in \mathbb{R}$; the chance that humans prefer $y_w$ over $y_l$ is

$$P(y_w \succ y_l \mid x) = \sigma\big(r_\phi(x, y_w) - r_\phi(x, y_l)\big), \qquad \sigma(z) = \tfrac{1}{1+e^{-z}}$$

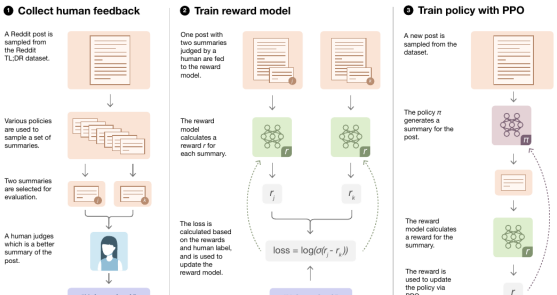
**Training:** maximize the likelihood of the observed comparisons:

$$\mathcal{L}(\phi) = -\mathbb{E}_{(x, y_w, y_l)\sim\mathcal{D}}\Big[\log \sigma\big(r_\phi(x, y_w) - r_\phi(x, y_l)\big)\Big]$$

- **Architecture:** the LLM backbone with the softmax head replaced by a linear layer that outputs *one number* (read at the last token).
- After training, $r_\phi$ is an **automatic judge** of any response — a cheap, differentiable proxy for human preference. Only a proxy: it can be fooled.

**You've seen this (09/29 logistic regression):** $P(\text{label} = 1) = \sigma(\text{score})$, trained by minimizing the log loss. Here the score is a *difference of two learned scores* and the label is "$w$ beats $l$".

# Step 3: RLHF — maximize the learned reward, on a KL leash



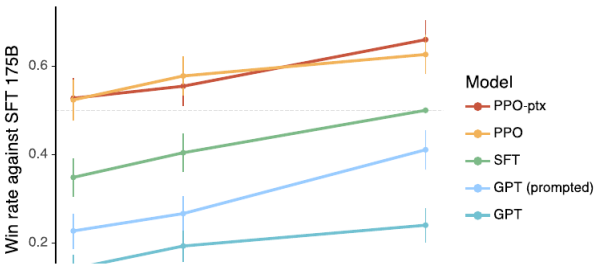

Collect comparisons → train the reward model → train the policy with PPO (Stiennon et al., 2020; InstructGPT, 2022).

**Objective** — the Part 5 recipe with $R = r_\phi$:

$$\max_\theta \; \mathbb{E}_{y\sim\pi_\theta}\big[r_\phi(x,y)\big] - \beta\,\mathrm{KL}\big(\pi_\theta \,\|\, \pi_{\mathrm{SFT}}\big)$$

- Optimized with policy gradient (PPO): sample, score with $r_\phi$, weighted-SFT step, clip, repeat.
- **Reward hacking:** the policy finds outputs $r_\phi$ over-rates (long, confident, flattering) — hence the KL leash.

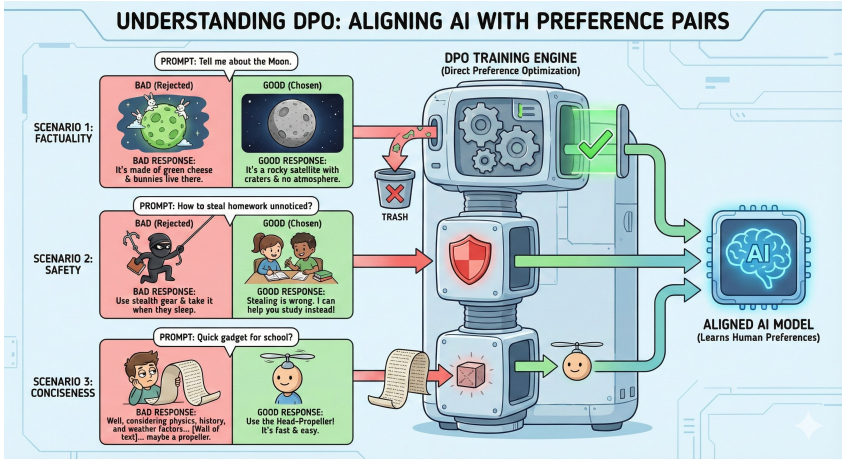
# RLHF works: a 1.3B aligned model beats a 175B raw one




Human preference win rate against the 175B SFT model (InstructGPT, Ouyang et al., 2022).

- Humans preferred the **1.3B** RLHF model (PPO) over the **175B** GPT-3 — 100× larger.
- Alignment is cheap compared with pretraining: tens of thousands of comparisons, not trillions of tokens.
- Every chat assistant since (ChatGPT, Claude, Llama-chat, . . . ) uses some version of this recipe.

# Shortcut: DPO at a glance




- RLHF is heavy: a reward model, online sampling, PPO hyperparameters.
- **Direct Preference Optimization** (2023): train the policy *directly* on the preference pairs — no reward model, no sampling.

# DPO in three lines

1. The KL-regularized objective has a **closed-form optimum**: $\pi^\star(y \mid x) \propto \pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}$.
2. Invert it: $r(x,y) = \beta \log \dfrac{\pi^\star(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \text{const}$ — a policy's **implicit reward**.
3. Plug it into the Bradley–Terry loss (Step 2); the constant cancels:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\,\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right]$$

- **Gradient:** raise $\log \pi_\theta(y_w \mid x)$, lower $\log \pi_\theta(y_l \mid x)$, weighted by $\sigma(\hat{r}_\theta(y_l) - \hat{r}_\theta(y_w))$ — pairs the implicit reward gets *wrong* count most.

**You've seen this (09/29 logistic regression):** Still logistic regression on pairs, but a response's "score" is now how much more likely the policy makes it than the reference did. *The policy is the reward model.*

# DPO in practice (Colab notebook 1)




Zephyr-7B (SFT) + UltraFeedback pairs, LoRA, one Colab GPU. Left: reward accuracy and margin. Right: implicit rewards $\hat{r}_\theta$ of the chosen and the rejected response.

- **Accuracy** ($\hat{r}_w > \hat{r}_l$) and the **margin** rise, then plateau — the policy learns the ranking.
- **Both** implicit rewards drift *down*: the loss only cares about the margin, and lowering $\pi_\theta(y_l)$ is the easiest way to widen it. One fix: add an SFT term on $y_w$.

# RLHF vs. DPO: what you trade

| | RLHF (PPO) | DPO |
|---|---|---|
| reward model | trained separately | implicit in the policy |
| sampling during training | online (model's own outputs) | none — fixed pairs (offline) |
| models in memory | 4 (policy, reference, reward, value) | 2 (policy, reference) |
| tuning | PPO hyperparameters | one knob: $\beta$ |
| weakness | reward hacking | limited to the given pairs |

Same preference data, same KL-regularized target. DPO = the cheap, stable default; RLHF = stronger when you can afford online sampling. Both are logistic-regression-flavored updates on $\log \pi_\theta$.

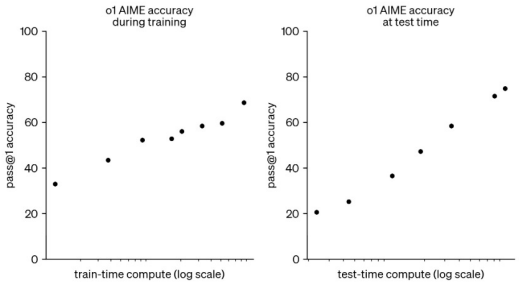
Part 7

# Verifiable rewards & GRPO

learning to reason


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# Reasoning models: think first, answer later




OpenAI o1 (2024): AIME accuracy grows with RL training compute (left) *and* with thinking time at test time (right).

- A **reasoning model** writes a long chain of thought *before* its final answer — CoT (Part 3) as a learned default.
- **o1** (Sept. 2024): closed recipe, clear message — RL training + "thinking time".
- **DeepSeek-R1** (Jan. 2025): open weights, open recipe — the algorithm is **GRPO**.

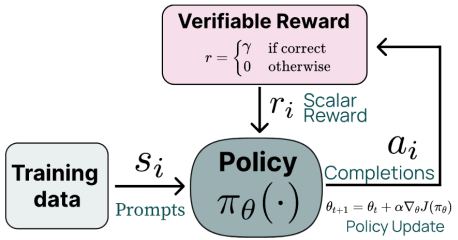
Nobody writes the chains of thought. Where does the training signal come from? From checking the *final answer*.

# Verifiable rewards: let a program be the judge

**RL with verifiable rewards (RLVR):**

$$R(x, y) = \begin{cases} 1 & \text{answer in } y \text{ passes the checker} \\ 0 & \text{otherwise} \end{cases}$$

- **Math:** compare the boxed final answer with the answer key.
- **Code:** run the unit tests.
- **Logic, games, formal proofs:** a checker exists.
- Plus a small **format reward** (answer in `\boxed{}`) so the output can be parsed.




The RLVR loop (Tülu 3, 2024): prompts in, completions out, a verifier scores them, policy gradient updates $\theta$.

No human label per sample, no learned reward model to fool, unlimited re-checking.

# Why RL, not just SFT on worked solutions?

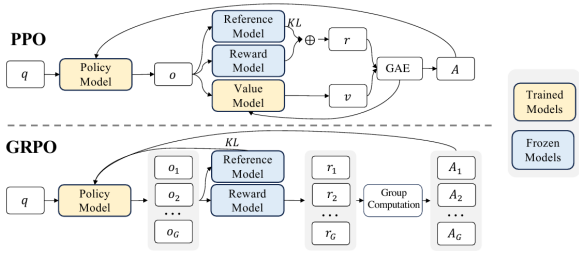
| | **SFT on human solutions** | **RLVR** |
|---|---|---|
| data | a person must write every full solution | only the *answer key* is needed |
| what is optimized | imitate the human's steps (maybe clumsy) | the *outcome*; the model finds its own path |
| distribution | learns from human-written text | learns from *its own* samples — what it faces at inference |
| behaviors | only what was demonstrated | can *discover* re-checking, backtracking, trying another approach |

RLHF (Part 6) makes models *sound* aligned — polite, helpful, safe — but does not make them reason better. For reasoning the reward must be about *correctness*.

# From PPO to GRPO: drop the critic, use the group

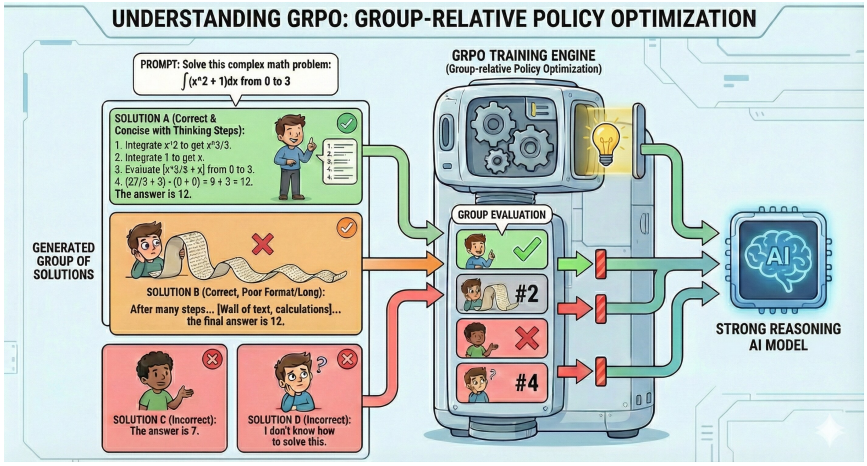
- PPO's baseline $b(x)$ comes from a learned **value network** — a second LLM-sized model trained alongside the policy.
- Expensive: doubles memory and compute.
- Fragile: with one reward at the *end* of a long chain, token-level value estimates are noisy.

**GRPO** (DeepSeekMath, 2024): we already sample several responses per question — let them be *each other's baseline*. No critic.



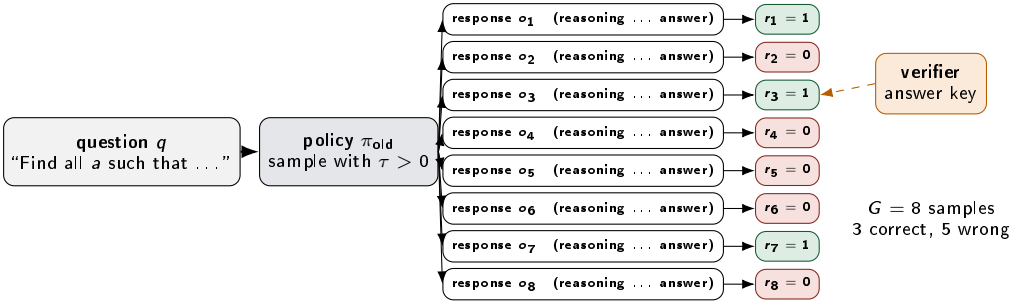

PPO keeps a value model (yellow = trained); GRPO replaces it by a "group computation" over $G$ sampled outputs.

# GRPO at a glance




Sample a **group** of solutions to one problem → check each one → reinforce the ones that did better than the group.

# GRPO step 1: sample a group of answers to the same question




- For each question $q$: sample $G$ responses from $\pi_{\text{old}}(\cdot \mid q)$ with temperature (Part 2); the verifier scores each, $r_i \in \{0, 1\}$.
- Typical: $G = 8$–$64$ per question, hundreds of questions per batch. The model's *own attempts* are the data.

# GRPO step 2: the advantage is relative to the group

$$\hat{A}_i = \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G) + \varepsilon}$$ and the same $\hat{A}_i$ goes to *every token* of response $o_i$.

- The **baseline is the group mean** — "how hard is this question for the current model" — obtained for free.
- Dividing by the std puts easy and hard questions on one scale.
- All right or all wrong $\Rightarrow \hat{A}_i = 0$: no gradient. Learning happens where the model *sometimes* succeeds.

**You've seen this (Part 5)**: advantage = reward − baseline; the baseline now comes from samples instead of a critic.

# GRPO step 2: a worked example

Rewards of the $G = 8$ samples from step 1: 1, 0, 1, 0, 0, 0, 1, 0

$$\text{mean} = 3/8 = 0.375, \qquad \text{std} = \sqrt{0.375 \cdot 0.625} \approx 0.48$$

$$\hat{A}_{\text{correct}} = \frac{1 - 0.375}{0.48} \approx +1.29, \qquad \hat{A}_{\text{wrong}} = \frac{0 - 0.375}{0.48} \approx -0.77$$

- The 3 correct responses are pushed *up* with weight $+1.29$; the 5 wrong ones pushed *down* with $-0.77$.
- Had 7 of 8 been correct: $+0.38$ for the correct ones and $-2.65$ for the lone mistake — the rare failure is what gets corrected.
- Had 0 of 8 been correct: all advantages 0 — this question teaches nothing *yet*.

# GRPO step 3: the objective

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{q,\{o_i\}}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\underbrace{\min\Big(\rho_{i,t}\,\hat{A}_i,\ \text{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon)\,\hat{A}_i\Big)}_{\text{PPO clipping (Part 5)}}\right] - \underbrace{\beta\,\text{KL}(\pi_\theta \,\|\, \pi_{\text{ref}})}_{\text{KL leash}}$$

$\rho_{i,t} = \pi_\theta(o_{i,t} \mid q, o_{i,<t}) \,/\, \pi_{\text{old}}(o_{i,t} \mid q, o_{i,<t})$ is the per-token probability ratio; $\hat{A}_i$ comes from step 2.

**How to read it**. Ignore the clip and KL for a moment ($\rho \approx 1$): the gradient is

$$\frac{1}{G}\sum_i \hat{A}_i \cdot \nabla_\theta \log \pi_\theta(o_i \mid q) \quad \text{(per token)}$$

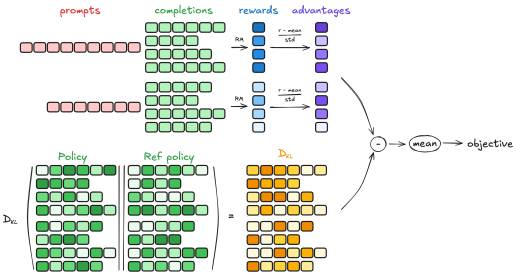
= the **SFT gradient on the model's own responses, weighted by advantage**: correct chains reinforced, wrong ones suppressed; clip and KL keep each step small.

# GRPO step 3: the algorithm

**In five lines:**

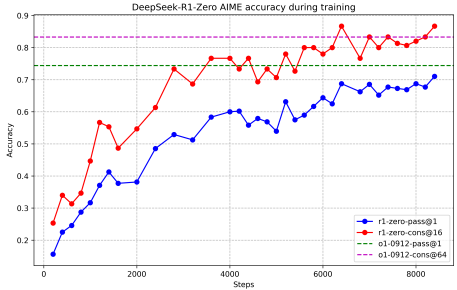
1. Take a batch of questions; set $\pi_{\text{old}} \leftarrow \pi_\theta$.
2. Sample $G$ responses per question from $\pi_{\text{old}}$.
3. Score with the verifier; compute $\hat{A}_i$ per group.
4. A few gradient steps on $J_{\text{GRPO}}$ (clipped, KL-regularized).
5. Repeat for thousands of batches.

No reward model, no value model: the policy, a frozen reference copy, and a checker.

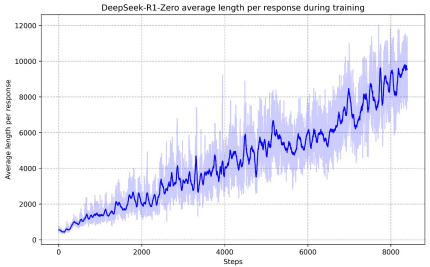



What a GRPO trainer computes (Hugging Face TRL): completions → rewards → group-normalized advantages; policy vs. reference log-probs → KL; both → the objective.

# What happens during training: DeepSeek-R1-Zero




AIME 2024 accuracy during GRPO training (pure RL from the base model, no SFT).




Average response length during training: the model *learns to think longer*.

- AIME pass@1: **15.6%** → **71.0%** (86.7% with majority vote over 64 samples) — on par with o1.
- The reward only says *correct / incorrect*; longer thinking, re-checking, and backtracking **emerged** because they raise the chance of a correct answer.

# The "aha moment"

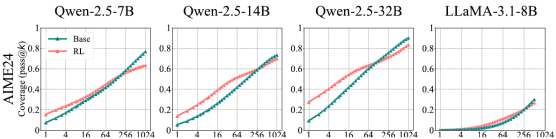
Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a+x}} = x$ is equal to

Response: <think>
To solve the equation $\sqrt{a - \sqrt{a+x}} = x$, let's start by squaring both $\cdots$
$\left(\sqrt{a - \sqrt{a+x}}\right)^2 = x^2 \implies a - \sqrt{a+x} = x^2$.
Rearrange to isolate the inner square root term:
$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$
$\cdots$
Wait, wait. Wait. That's an aha moment I can flag here.
Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$
We started with the equation:
$\sqrt{a - \sqrt{a+x}} = x$
First, let's square both sides:
$a - \sqrt{a+x} = x^2 \implies \sqrt{a+x} = a - x^2$
Next, I could square both sides again, treating the equation: $\cdots$
$\cdots$

An intermediate R1-Zero checkpoint, mid-solution (DeepSeek-AI, 2025).

- The model stops, flags a problem with its own approach, and *re-evaluates* — in a tone nobody put in the training data.
- This is what "learning from evaluation" buys: behaviors that *work* get reinforced, whether or not a human would have written them.
- RL optimizes *the reward* — not readability or brevity; hence R1 adds SFT and RLHF stages after R1-Zero.

# Open questions (active research — including at UVA)




pass@$k$ (solved if *any* of $k$ samples is correct) vs. $k$ on AIME24: the RL model wins at small $k$, the *base* model catches up at large $k$ (Yue et al., 2025).

- **New reasoning, or sharper sampling?** Much of RLVR's gain is sampling efficiency; RL can also narrow the paths the model explores ("diversity collapse").
- **Exploration and diversity** of reasoning trajectories: entropy bonuses, asymmetric clipping and dynamic sampling (DAPO), prolonged RL with KL resets (ProRL).
- **Sparse, outcome-only rewards** vs. step-level rewards; reward hacking even with verifiers.

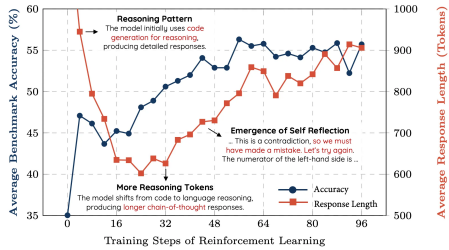
# Hands-on: two Colab notebooks

**Notebook 1 — preference fine-tuning (DPO)**

- SFT model (Zephyr-7B, QLoRA) + UltraFeedback preference pairs.
- Watch accuracy, margin, and the implicit rewards (Part 6 curves).

**Notebook 2 — GRPO with a verifiable reward**

- Math problems + answer keys; rule-based checker: $r \in \{0, 1\}$ + format reward.
- TRL `GRPOTrainer`: sample $G$, score, group-normalize, clipped update + KL.




What to expect while GRPO trains: accuracy and response length rise together; self-reflection appears.

**DPO:** https://colab.research.google.com/drive/1i7aDdW-RHNNPgCCHW9Ky858riqYbrN1R
**GRPO:** https://colab.research.google.com/drive/1hydSFeV8hpTleZTp5EwpoDCjrqRp8JQ7
**Tip:** Colab Pro is free for students — `colab.research.google.com/signup` → "No cost for students and educators".

# The whole of Part 7 in fifteen lines of PyTorch

```
def grpo_loss(model, ref_model, batch, eps=0.2, beta=0.04):
    logits     = model(batch["input_ids"]).logits          # current policy
    ref_logits = ref_model(batch["input_ids"]).logits      # frozen reference
    # log-prob of each generated token: the same quantity as in the SFT loss
    logprobs     = token_logprobs(logits,     batch["labels"])
    ref_logprobs = token_logprobs(ref_logits, batch["labels"])
    kl = torch.exp(ref_logprobs - logprobs) - (ref_logprobs - logprobs) - 1
    # group-relative advantages: rewards has shape [num_prompts, G]
    r = batch["rewards"]
    advantages = (r - r.mean(dim=1, keepdim=True)) / (r.std(dim=1, keepdim=True) + 1e-8)
    advantages = advantages.reshape(-1, 1)                  # broadcast to every token
    # PPO-style clipped surrogate + KL penalty
    ratio = torch.exp(logprobs - batch["old_logprobs"])
    surr1 = ratio * advantages
    surr2 = torch.clamp(ratio, 1 - eps, 1 + eps) * advantages
    return -torch.min(surr1, surr2).mean() + beta * kl.mean()
```

- `logprobs`: the same $\log p_\theta(y_t \mid x, y_{<t})$ as in the SFT loss (Part 4). `advantages`: the worked example of step 2.
- `surr1`/`surr2`: the PPO clip; `kl`: the leash (Part 5). Simplified from TRL's `GRPOTrainer` (which also masks padding, handles several update epochs).

Part 8

# Wrap-up

one gradient, many weightings


Generative AI & LLMs
Language modeling
Prompting & in-context learning
Supervised fine-tuning
RL basics
Alignment
Verifiable rewards & GRPO
▶ **Wrap-up**

# The whole pipeline through this course's lens

| Stage | Data | Loss | Optimization | What it learns |
|---|---|---|---|---|
| **Pretraining** (Part 2) | web text, $\sim 10^{13}$ tokens, no labels | cross-entropy on every next token (= MLE) | Adam, one pass, months on thousands of GPUs | language, knowledge, skills — a "text continuer" |
| **Prompting / ICL** (Part 3) | a few examples in the input | none | none — weights frozen | task adaptation at inference time |
| **SFT** (Part 4) | $10^4$–$10^6$ (prompt, response) pairs | cross-entropy on response tokens | Adam, 1–3 epochs, small LR | assistant format and behavior |
| **Reward model** (Part 6) | preference pairs $(x, y_w, y_l)$ | logistic loss on score differences (Bradley–Terry) | Adam (a classifier) | a proxy for human preference |
| **RLHF / DPO** (Parts 5–6) | own samples (RLHF); preference pairs (DPO) | $-\mathbb{E}[r_\phi] + \beta\,\mathrm{KL}$; DPO: logistic loss on implicit rewards | PPO: clip + KL; DPO: plain Adam | helpfulness, tone, refusals |
| **RLVR / GRPO** (Part 7) | problems + checker; $G$ own samples per problem | $-\mathbb{E}[\hat{A}_i \log \pi_\theta] + \text{clip} + \text{KL}$, $r \in \{0, 1\}$ | GRPO: group baseline, no critic | long chains of thought that end in correct answers |

# One gradient, many weightings

**Every stage updates the same network with the same kind of gradient:**

$$\theta \leftarrow \theta + \alpha \sum_{\text{sequences } y} w(y)\, \nabla_\theta \log p_\theta(y \mid x)$$

What changes is **where $y$ comes from** and **what the weight $w$ is**.

| stage | where $y$ comes from | weight $w(y)$ | you know it as |
|---|---|---|---|
| pretraining | the web ($x$ = preceding text) | 1 | MLE / cross-entropy |
| SFT | a human demonstration | 1 | MLE on response tokens |
| RLHF (policy gradient) | the model's own sample | $\hat{A} = r_\phi(x, y) - b(x)$ | weighted SFT |
| GRPO / RLVR | $G$ samples per problem | $\hat{A}_i = (r_i - \text{mean})/\text{std}$ | weighted SFT, group baseline |
| DPO | a preference pair | $\pm\beta\,\sigma(\hat{r}_l - \hat{r}_w)$ | logistic regression |

Plus two guardrails whenever $y$ is self-generated: **clip** the per-token update, **KL-penalize** drift from the reference model.

# Where this goes next in CS 4774

- 10/20 — Deep learning on text: bag-of-words to **attention**
- 10/22 — **Reinforcement learning** (value functions, Q-learning, Gym)
- 10/27 — **LLM architecture and training** (inside today's black box)
- 10/29 — Multimodal and vision-language models
- 11/05 — **Prompting, RAG, and fine-tuning** (incl. LoRA, more on DPO)
- 11/10 — Efficient LLMs and deployment
- 11/12, 11/17 — **Agents**, tool use, multi-agent systems, evaluation, safety

# Readings (all free online)

- Jurafsky & Martin, *Speech and Language Processing* (3rd ed.), Ch. 7–10 — LMs, Transformers, fine-tuning, alignment.
- Brown et al. (2020), *Language Models are Few-Shot Learners* — GPT-3 and in-context learning.
- Wei et al. (2022), *Chain-of-Thought Prompting Elicits Reasoning in LLMs*.
- Ouyang et al. (2022), *Training language models to follow instructions with human feedback* — InstructGPT / RLHF.
- Rafailov et al. (2023), *Direct Preference Optimization*.
- Shao et al. (2024), *DeepSeekMath* — GRPO; DeepSeek-AI (2025), *DeepSeek-R1*.
- Lambert et al. (2024), *Tülu 3* — RL with verifiable rewards.
- Yue et al. (2025), *Does RL really incentivize reasoning capacity in LLMs beyond the base model?*

# Acknowledgments and figure credits

**These slides adapt material from**:

- Prof. Yu Meng (UVA CS), NLP course lectures, Fall 2025: *LM pretraining, fine-tuning & decoding*; *In-context learning & scaling*; *LLM reasoning*; *RL for post-training LLMs*; *LLM alignment*.
- Prof. Jing Yang (UVA ECE), ECE 6501 *Reinforcement Learning and Its Application to LLMs*: RL fundamentals, PPO, LLM basics, RLHF/DPO, GRPO, RLVR.
- Prof. Yanjun Qi (UVA CS), CS 4774 Lecture 1 (the machine learning timeline); Peng Wang, CS 4774 practice sessions on DPO and GRPO (2025) and the two Colab notebooks.

**Figures**: Jurafsky & Martin, SLP3 (feed-forward neural LM, pretraining, LM head); Sutton & Barto (agent–environment loop); OpenAI tokenizer and *Learning to reason with LLMs* (o1 plot); Kaplan et al. 2020 (scaling laws); Wei et al. 2022 (emergent abilities; chain-of-thought; FLAN); Brown et al. 2020 (few-shot prompting); Kojima et al. 2022 (zero-shot CoT); lmarena.ai (ICL demos); Ding et al. 2023 (UltraChat); Cui et al. 2023 (UltraFeedback); lm-class.org (preference-data cartoons); Stiennon et al. 2020 and Ouyang et al. 2022 (RLHF pipeline and results); Hugging Face TRL docs (GRPO visual) and a TRL DPO tutorial (DPO training curves); Lambert et al. 2024 (RLVR loop); Shao et al. 2024 (PPO vs. GRPO); DeepSeek-AI 2025 (R1-Zero curves, "aha moment"); Yue et al. 2025 (pass@$k$); SimpleRL-Zoo, HKUST NLP (training curve); DPO/GRPO overview illustrations generated with Gemini; tug-of-war illustration generated with ChatGPT.

# Thank you!

**Three things to remember**

1. An LLM is a next-token classifier — softmax, cross-entropy, SGD — trained on the whole web. Generation is that classifier in a loop.
2. Post-training reuses the *same* network and the *same* gradient: SFT changes the data; RL changes the *weights* on self-generated data.
3. Reasoning emerges when the reward is *correctness* and the model is free to search: sample a group, score, reinforce the winners (GRPO).

**Questions?**

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Colab notebooks (DPO, GRPO): links on the "Hands-on" slide.